EXPERIMENT 1

AIM

To develop a Python program to:

1. Represent student grade distribution using pie and bar charts, identify maximum and minimum grades, and calculate the combined percentage of A and S grades.
2. Detect objects in an image using the YOLO machine learning algorithm
3. Compare Logistic Regression and Random Forest algorithms for heart disease prediction using classification accuracy.

ALGORITHM

- Start the program.
- Import the required Python libraries such as Matplotlib, Pandas, Scikit-learn, and YOLO.
- Store the grades and corresponding number of students in lists.
- Draw a pie chart to display the percentage distribution of students across all grades.
- Draw a bar chart to compare the number of students in each grade and display the values above the bars.
- Identify the grades with the maximum and minimum number of students.
- Calculate and display the combined percentage of students who secured grades A and S.
- Upload an image and load the pretrained YOLO model.
- Detect objects in the image and display the detected objects with bounding boxes.
- Load the heart disease dataset and separate the input features and target variable.
- Split the dataset into training and testing sets.
- Train the Logistic Regression and Random Forest classification algorithms.
- Predict the test results using both algorithms.
- Calculate and display the accuracy of each algorithm for comparison.
- Stop the program.

CODE:

```
# Install required libraries

!pip install -q matplotlib pandas scikit-learn ultralytics



import matplotlib.pyplot as plt
```

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import make_pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
from ultralytics import YOLO
from google.colab import files

# 1. GRADE DISTRIBUTION
grades = ['F', 'E', 'D', 'C', 'B', 'A', 'S']
students = [3, 4, 5, 8, 9, 7, 4]

plt.figure(figsize=(6, 6))
plt.pie(students, labels=grades, autopct='%1.1f%%', startangle=90)
plt.title("Grade Percentage Distribution")
plt.legend(grades, title="Grades")
plt.show()

plt.figure(figsize=(7, 5))
bars = plt.bar(grades, students, color='skyblue')
plt.xlabel("Grades")
plt.ylabel("Number of Students")
plt.title("Students in Each Grade")
```

```
plt.bar_label(bars)
plt.show()

print("Maximum students:", grades[students.index(max(students))], max(students))
print("Minimum students:", grades[students.index(min(students))], min(students))
print("A and S combined percentage:",
      (students[5] + students[6]) / sum(students) * 100, "%")



# 2. OBJECT DETECTION USING YOLO
print("\nUpload an image for object detection:")
uploaded = files.upload()
filename = next(iter(uploaded))

model = YOLO('yolov8n.pt')
results = model(filename)
image = results[0].plot()

plt.figure(figsize=(8, 6))
plt.imshow(image[:, :, ::-1])
plt.title("Object Detection Using YOLO")
plt.axis('off')
plt.show()
```

```
# 3. HEART DISEASE PREDICTION
url = "https://raw.githubusercontent.com/plotly/datasets/master/heart.csv"
df = pd.read_csv(url)

X = df.drop("target", axis=1)
y = df["target"]

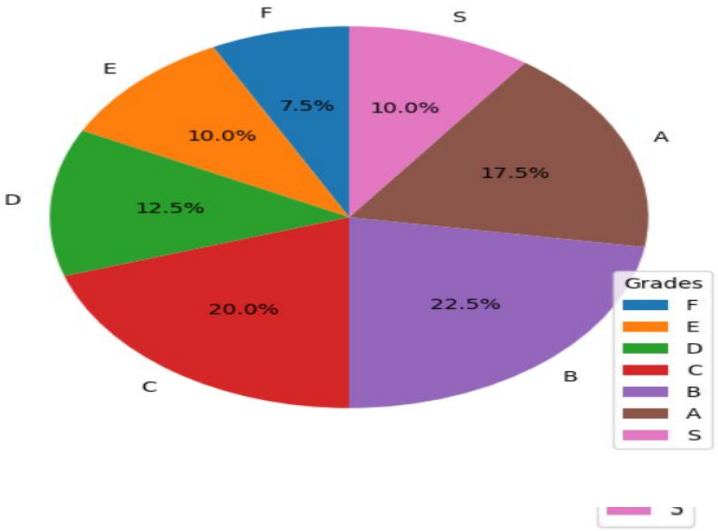
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

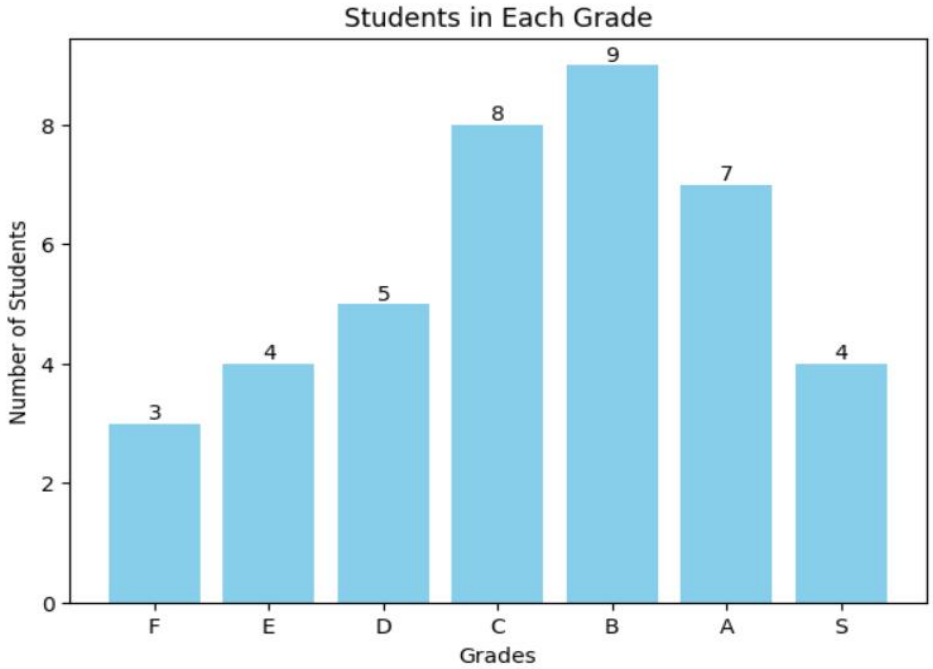
models = {
    "Logistic Regression": make_pipeline(
        StandardScaler(), LogisticRegression(max_iter=1000)
    ),
    "Random Forest": RandomForestClassifier(
        n_estimators=100, random_state=42
    )
}

print("\nHeart Disease Algorithm Comparison:")
for name, model in models.items():
    model.fit(X_train, y_train)
    prediction = model.predict(X_test)
    accuracy = accuracy_score(y_test, prediction)
```

print(name, "Accuracy:", round(accuracy * 100, 2), "%")

OUTPUT:






```
Maximum students: B 9
Minimum students: F 3
A and S combined percentage: 27.500000000000004 %
```